

K.A.S.I. Ranaweera

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Profile

Software Engineering undergraduate specializing in AI/ML and full-stack development with practical experience in computer vision, data processing and building intelligent applications. Experienced in generative AI and RAG architectures, including an AI-powered education platform that adapts to individual learning styles, as well as a range of other projects. Active technical leader with experience in university project management and advisory teams.

Experience

OREL IT

Associate Image Processor

Maharagama, Sri Lanka

Nov 2021 – Jun 2022

- Contributed to international autonomous-driving projects through image and video data preparation.
- Processed and annotated 10,000+ frames from road video clips, achieving 98% accuracy rate for autonomous vehicle training datasets.
- Collaborated with cross-functional teams to meet project deadlines while strictly adhering to labeling guidelines and quality standards.
- Gained practical experience in computer vision data preprocessing, annotation tools, and dataset management.

Education

Sri Lanka Technology Campus, Padukka

Jun 2022 – present

- BSc (Hons) Software Engineering
 - Current GPA: 3.5/4.0
 - Expected Graduation - July 2026

Mahinda Rajapaksha College, Homagama

Jan 2012 – Oct 2020

- Advanced Level – Results: C, S, S
- Ordinary Level – Results: A-7, B-2

Courses and Certifications

- **IBM AI Engineering Specialization** – IBM (Coursera Platform) [View Certificate](#)
- **AI/ML Engineer - Stage 1** – SLIIT [View Certificate](#)
- **How to create a Jira SCRUM project** – Coursera Platform [View Certificate](#)
- **Cybersecurity Essentials Course & Exam** – Cisco Networking Academy [View Certificate](#)
- **Introduction to Cybersecurity Course & Exam** – Cisco Networking Academy [View Certificate](#)
- **TensorFlow Developer Professional Certificate** - DeepLearning.AI (*Currently Learning*)
- **Full Stack Software Engineering Course** – CODL - University of Moratuwa (*Currently Learning*)
- **ML, DL and Generative AI Course** – Coursera & DeepLearning.ai Platforms (*Currently Learning*)

Technical Projects

EduWingz (Final Year Project) [View on GitHub](#)

- Developed adaptive learning platform using generative AI and RAG architecture to identify student learning styles, generate personalized content, and dynamically adjust teaching methods
- Tools Used: React JS, Django, RAG, LangChain, Deepseek, Python

Whisper AI-Psychiatric – Built AI-powered mental health chatbot with voice interaction, fine-tuned Gemma-3 model for psychological guidance, RAG pipeline for context-aware responses, and crisis detection system [View on GitHub](#)

Z-Score Uni Finder – Developed ML recommendation system analyzing Z-scores and UGC cutoff data (2020-2024), achieving 92% prediction accuracy and published dataset receiving 100+ downloads. Published Dataset to Kaggle & Hugging Face Platforms [Dataset Link](#) [View on GitHub](#)

GNN on ogbn-arxiv – Implemented Graph Neural Network for node classification on arXiv citation dataset, experimenting with multiple GNN architectures (GCN, GraphSAGE, GAT) achieving 71% test accuracy [View on GitHub](#)

Road Accident Prediction – Built ML model predicting accident likelihood with Flask chatbot for real-time safety recommendations, achieving 87% prediction accuracy using Random Forest classifier [View on GitHub](#)

React Native Mobile App – Built Task Manager App while learning React-Native via the YouTube Crash Course

Brainbox Quiz App – Developed full-stack quiz system with React.js frontend and Django backend for interactive learning and assessment. Frontend: [View on GitHub](#) Backend: [View on GitHub](#)

Academics & Notable Projects – Platform for managing SLTC academic resources, schedules, and student tools.

- Smart Fan Control System with Gesture Recognition – IoT-based system integrating sensors and machine learning gesture recognition for automated fan control. [View on GitHub](#)
- AR Space Education Application – An immersive Augmented Reality (AR) educational application focused on solar system exploration and space phenomena. [View on GitHub](#)

Technical Skills

Programming Languages – Python, JavaScript, TypeScript, Java, SQL, Dart

AI / ML / DL – TensorFlow, PyTorch, Scikit-learn, Keras, Transformers, Hugging Face, LangChain, RAG, Vector Databases (Chroma/FAISS), Pandas, NumPy, Streamlit

Web Development – React.js, Next.js, Django, Flask, FastAPI, Node.js, REST APIs, Tailwind CSS, Material UI

Tools & Platforms – Git/GitHub, Docker, Jira, Google Colab, Kaggle, AWS, Vercel, Postman, VS Code

Leadership & Community Involvement

- **SEDS SLTC** | *Vice Chairman (2024-2025), Assistant Project Manager (2023-2024), Member (2022-2023)* – Key Projects: Project Manager for Stargaze '23; Advisor for Stargaze '24. [View Details](#)
- **IEEE Industry Applications Society of SLTC** | *Assistant Webmaster (2024-2025), Membership Development Head (2023-2024), Member (2022-2023)* – Key Projects: Co-Project Manager for Venturify'24 and InnovMind V2.0; Design Team Member for Technomind. [View Details](#)
- **NASA Space Apps Colombo** | *Event & Logistics Team Member* – Coordinated logistics, venue setup, prize-giving, and participant support for NASA's global hackathon. [View Details](#)
- **Media Unit of SLTC** | *Program Head (OC) of Master Designer v2.0* – Directed program flow for national graphic design competition; organized workshops and publicity. [View Details](#)
- **Mahinda Rajapaksha College** | *Deputy Head Prefect (2020), House Captain (2019), Member, School Cadet Platoon (2015-2019)* – During school years, actively participated in leadership roles, extracurricular activities, and sports, developing teamwork, discipline, and organizational skills. [View Details](#)

Achievements & Recognitions

- **Codemanía Hackathon**: 4th Place (Codemanía v5.0 - 2023) and 7th Place (Codemanía v3.0 - 2025) [View Certificate](#)
- **SLTC Venture Program**: Selected for the Final Round and awarded a Certificate of Recognition
- **Rocketry Design & Modeling Competition**: Earned Outstanding Performance Certification for design accuracy

Additional Information

- **Language**: Sinhala, English
- **Abilities & Skills**: Creativity, Teamwork, Responsibility Management, Critical thinking
- **Interests**: AI research, full-stack web development, open-source contribution, community mentorship